

dataset covers more than 280,000 U.S. firms, more than 160 million employment spells from roughly 65 million unique workers between 2015 and 2025, and almost 200 million job postings. A key advantage of the data is the standardized seniority classification assigned to each position by Revelio’s algorithm, which enables us to track junior and senior employment within firms over time.⁴

We identify GenAI adoption by detecting job postings that explicitly recruit “GenAI integrator” roles. The method follows the approach of [Hampole et al. \(2025\)](#) and proceeds in two steps: first flagging postings with GenAI-related keywords, then using a large language model (LLM) to determine whether the posting reflects a genuine integrator position—one dedicated to implementing or operating GenAI technology in the firm’s workflow. A firm is classified as an adopter if it has posted at least one such vacancy, thereby capturing firms that have actively initiated the integration of GenAI into their operations.⁵

By this definition, 10,433 firms in our sample adopted GenAI by March 2025. Although adopters represent only about 3.7 percent of the 281,000 firms in our sample, they are substantially larger on average and account for roughly 16 percent of total employment in our dataset. Our analysis shows that adoption of GenAI was minimal and relatively stable prior to 2023, but accelerated sharply thereafter, with a surge of new firms posting integrator roles following the release of ChatGPT in late 2022. We further validate this firm-level adoption measure using worker-side data. Applying a list of GenAI keywords to position descriptions, we find a sharp differential increase in AI-related mentions at adopting firms after the launch of ChatGPT, supporting the interpretation that our adoption measure captures genuine changes in workplace technology use.

The main empirical exercise of the paper is to test the framework’s central implication: whether GenAI adoption disproportionately reduces junior employment relative

⁴We validate the data in Appendices [A.1](#) and [A.2](#). First, we show that the composition of the LinkedIn sample relative to QCEW, OEWS, and QWI benchmarks remains stable over the GenAI diffusion period. Second, we test for differential reclassification: the concern that adopting firms changed how positions were classified as junior after 2022. Using both job-title keywords and workers’ actual labor-market experience, we find that the content of each seniority group evolves similarly at adopting and non-adopting firms, with no break around the ChatGPT launch. This indicates that the decline in junior employment is unlikely to be driven by changes in seniority classification.

⁵Our concept of a GenAI integrator is closely related to the “robot integrator” in [Acemoglu and Restrepo \(2020\)](#), defined as “companies that install, program, and maintain robots.” They use the presence of robot integrators as a proxy for local robot adoption and show that commuting zones more exposed to robots have a higher number of integrators.

to senior employment. Before turning to this aggregate evidence, we first examine the framework’s underlying mechanisms. Using an LLM, we map a representative sample of our job postings to standardized O*NET tasks and show that AI-exposed tasks become disproportionately less likely to appear in junior task bundles at adopting firms after 2023, while no comparable pattern emerges for seniors. This provides evidence consistent with the displacement channel. By contrast, tasks that were ex ante concentrated among seniors do not systematically migrate into junior task bundles, offering little support for the reallocation channel. We complement these tests with external task-level data from the Anthropic Economic Index (Appel et al., 2026), documenting that real-world GenAI use is more likely to take the form of automation for tasks characteristic of junior occupations, whereas usage in senior occupations is more often augmentative, consistent with the framework’s assumption that junior tasks are more susceptible to displacement.

We then turn to the firm-level employment response. We first compare junior and senior employment at adopting and non-adopting firms. From 2018 through 2022, junior employment evolves similarly in the two groups. Beginning in late 2022, however, junior employment at adopters diverges sharply downward relative to non-adopters, while senior employment shows no comparable break. To directly assess the seniority-biased effect, we estimate a triple-difference specification that compares junior versus senior employment within adopting firms relative to non-adopters. This design includes firm-by-time fixed effects, which absorb shocks or trajectories specific to a given firm in a given period, and industry-by-time-by-seniority fixed effects, which account for sector-level dynamics that differentially affect juniors and seniors. The coefficients are flat through 2022Q4 and then decline sharply from 2023Q1 onward, reaching roughly a 9 percent decline in junior employment relative to senior employment after six quarters.

We next examine which types of jobs drive our results, focusing on heterogeneity by occupational exposure to GenAI using the widely used exposure measure of Eloundou et al. (2024). Specifically, we estimate a triple-difference specification that compares employment changes in high- versus low-exposure occupations within adopting and non-adopting firms, separately for junior and senior workers. Through the end of 2022, the estimated coefficients for juniors and seniors follow nearly identical trajectories. Beginning in 2023Q1, however, the two series diverge sharply. Junior employment in high-exposure occupations declines by roughly 7 percent relative to low-exposure occupations over the

subsequent two years, while the corresponding senior series continues to rise. This pattern is consistent with GenAI substituting for exposed junior tasks while leaving demand for senior workers in those occupations largely unaffected.⁶

We complement these analyses with a staggered event study that traces employment dynamics around firms' GenAI adoption date, proxied by the first period in which the firm posts a GenAI integrator role. This design helps distinguish adoption effects from broader time-specific shocks by exploiting variation in the timing of adoption across firms. However, it is sensitive to measurement error in the adoption timing proxy—for instance, if firms begin using GenAI before, or only several periods after, posting for an integrator role. The estimates reveal no evidence of differential pre-trends: employment evolves similarly in adopting and non-adopting firms throughout the eight quarters preceding adoption. However, roughly two quarters after adoption, junior employment in adopting firms begins to decline, reaching an 8 percent reduction after eight quarters. The absence of significant pre-trends provides additional reassurance that these post-adoption declines are unlikely to be driven by confounding shocks.

We next address potential endogeneity concerns. Adopting and non-adopting firms differ systematically, which could bias our estimates; for example, if adopting firms are more responsive to other shocks affecting employment. One concern is that GenAI adopters may have been more sensitive to the 2022–2023 monetary tightening cycle. Several pieces of evidence suggest that this mechanism does not drive our findings: including industry-by-seniority-by-time fixed effects leaves the results essentially unchanged; the event-study estimates show no pre-adoption trends; and a formal test based on the high-frequency identified monetary-policy surprises of [Jarocinski and Karadi \(2020\)](#), following [Ottonello and Winberry \(2020\)](#), finds no evidence that adopter firms are more responsive to monetary shocks. This result is also consistent with the literature documenting that larger firms—which are more likely to adopt GenAI—tend to be less sensitive to monetary policy ([Chodorow-Reich, 2014](#); [Gertler and Gilchrist, 1994](#)).

We also consider three interrelated concerns arising from the post-pandemic labor market: the hiring boom and subsequent correction in the technology sector, the expan-

⁶Supplemental Appendix [A.3](#) reports a complementary heterogeneity exercise by juniors' educational background. Measuring school quality by the prestige of juniors' alma mater, we find a U-shaped pattern: juniors from mid-tier universities experienced the largest relative declines, while those from the most and least prestigious institutions saw smaller reductions.

sion of remote work, and supply-side shocks to junior labor. We find little support for any of these explanations. The absence of pre-trends and the inclusion of industry-by-seniority-by-time fixed effects suggest that post-COVID sectoral dynamics do not drive our results. Re-estimating our triple-difference specification separately using employment in high- and low-remote-suitability positions yields similar declines in junior employment, suggesting that the results are not driven by shifts in remote work. Finally, job postings decline sharply at adopting firms after 2023, indicating that the reduction in junior employment reflects weaker labor demand rather than constraints on labor supply. Taken together, this evidence suggests that, although these factors are important for broader labor-market trends, they are unlikely to explain the specific effect identified by our empirical design.

We next examine the mechanisms underlying the decline in junior employment. Using our linked employer-employee data, we decompose workforce dynamics into inflows (hires), outflows (separations), and internal promotions. We find that the decline in junior employment at adopting firms is driven primarily by a substantial reduction in hiring. Separation rates for juniors in adopting firms also decreased relative to non-adopters, but the reduction in hiring was considerably larger, leading to a net decline in junior positions. Promotion rates rose by a small but statistically significant amount, though the magnitude is economically small relative to the hiring channel.

Taken together, these results indicate that GenAI adoption is associated with reduced junior employment. However, the relatively early and pronounced declines—starting to emerge soon after the release of ChatGPT—may seem surprising, as automation impacts typically materialize more slowly. This suggests that the decline may not reflect immediate task automation but rather forward-looking adjustments by firms: the rapid diffusion of GenAI may have shifted firms’ expectations, leading them to scale back hiring for roles they predict will be automated in the near future, consistent with broader evidence on asymmetric employment adjustment (Ilut et al., 2018). We formalize this mechanism in a simple dynamic model in which expectations of *future* automation—combined with labor-adjustment costs—lead firms to reduce hiring *today* (Supplemental Appendix A.5). Several patterns are consistent with this interpretation: GenAI adoption accelerated sharply in early 2023; mentions of “AI” in firms’ earnings calls began rising in 2022Q4; prominent U.S. companies publicly adjusted hiring practices in early 2023,

attributing these changes to GenAI; and the junior employment decline is driven primarily by slower hiring rather than increased separations. While these patterns align with forward-looking adjustments, our data do not allow us to test this channel directly. One potential implication is that if firms overestimate the near-term automation capabilities of GenAI, the resulting decline in junior employment may ultimately prove temporary.

We conclude by noting several limitations. Our adoption definition captures deliberate integration of GenAI into firm workflows, but does not account for “silent” adoption, such as employees using GenAI tools without firm integration. Therefore, our definition can be thought of as conservative: more likely to misclassify adopters as non-adopters (false negatives) than the reverse (false positives). Such misclassification, by assigning treated firms to the control group, is likely to attenuate estimates toward zero. However, silent adoption may have distinct implications for employment dynamics from the deliberate, firm-level adoption we capture.⁷ An additional caveat is that, as with any empirical analysis based on cross-sectional variation, our results do not necessarily generalize to the aggregate economy without additional assumptions, due to the “missing intercept problem” (see [Acemoglu and Restrepo, 2020](#); [Wolf, 2023](#); [Moll and Hanney, 2025](#)).

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 develops the conceptual framework. Section 4 describes the data and descriptive patterns. Section 5 presents our empirical strategy and main findings. Section 6 concludes.

2 Related Literature

Our study relates to three main strands of the literature: (i) skill-biased technological change, (ii) firm-level adoption of AI technologies, and (iii) the emerging evidence on the labor-market impacts of GenAI.

Skill-Biased Technological Change: The classic literature on skill-biased technological change (SBTC) shows that computers and automation have historically displaced

⁷We partly address this concern in Supplemental Appendix A.4 by constructing an augmented adoption measure that combines integrator postings with worker position descriptions classified as integrator roles; the headline result is robust to this alternative definition.